Richer-count first-order consolidation addendum

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Abstract

A binary first-order transition table already exhibits a logarithmic price for unknown-budget prediction from eight triple counts and boundary windows. In the original length-$M+2$, protected-terminal-pair experiment, the finite factor is between $2^{-33}\ell$ and $2^{17}\ell$, where $\ell = 1 + \ln((M+2)/r)$. The upper allows arbitrary initial pair laws and requires no mixing certificate. The key interface discards the first symbol, reconstructs the corrupted suffix counts, and applies one suffix-only predictor. This addendum consolidates accepted local results and their proof dependencies; it is not a standalone complete proof supplement or a replacement manuscript. Current integration status is recorded separately in `STATUS.md` and its receipt.

1 One common experiment

Fix integers $M \geq 64$, $2 \leq r \leq M+2$. A clean word is $X_0, \ldots, X_{M+1} \in \{0,1\}$. Its initial *pair* has an arbitrary hidden law μ. A hidden first-order table T governs exactly the transitions from X_1 onward:

$$\Pr(X_{i+1} = b \mid X_0, \ldots, X_i) = T(X_i, b), \qquad 1 \leq i \leq M.$$

The initial edge need not follow T. There are M triple departures but $M+1$ ordinary edges. For the stationary subclass, $\mu(a,b) = \pi(a)T(a,b)$ for a stationary symbol law π.

A source-independent offline kernel K maps the word to Z and, on *every* input and outcome, changes at most integer $k \in \{0, \ldots, M\}$ coordinates among $0, \ldots, M-1$. The final two symbols are protected: $(Z_M, Z_{M+1}) = (X_M, X_{M+1})$. The kernel is a fixed word-to-word map, not a map receiving the true law. Predictor randomness is independent of attack randomness.

The observation is

$$\mathcal{T}_r(Z) = \big((H_{abc}(Z))_{a,b,c\in\{0,1\}},\ Z_{0:r-1},\ Z_{M+2-r:M+1}\big), \quad H_{abc}(Z) = \sum_{i=0}^{M-1} \mathbf{1}\{Z_i Z_{i+1} Z_{i+2} = abc\}.$$

Windows may overlap; all counts and supplied symbols are exact. The predictor outputs a next-symbol distribution. Its unconditional unsquared risk is

$$R(A; T, \mu, K) = \mathbb{E}\ \mathrm{TV}\left(A(\mathcal{T}_r(Z)), T(X_{M+1}, \cdot)\right).$$

This equals the actual-terminal-pair prediction loss after lifting both distributions to pairs whose first coordinate is X_{M+1}. No independent test query is substituted.

For each fixed (M, r), one map A must serve *all* budgets. With $g_M(k) = \max\{M^{-1/2}, k/M\}$ define

$$\Lambda_\star(M,r) = \inf_{A\ k\text{-blind}} \sup_{\substack{k\in\{0,\ldots,M\},\ T,\mu,K \\ \text{qualifying } \star \text{ experiments}}} \frac{R(A;T,\mu,K)}{g_M(k)}, \qquad \star \in \{\mathrm{stat}, \mathrm{all}\}.$$

Maps may be randomized and depend on M, r, not on the hidden table, law or budget. No infimum or worst case is assumed attained.

2 Finite characterization and exact benchmark

Theorem 1 (First-order richer-count window price)**.** *In the common experiment, for all first-order tables without a mixing assumption,*

$$\boxed{2^{-33}\ell \;\le\; \Lambda_{\mathrm{stat}}(M,r) \;\le\; \Lambda_{\mathrm{all}}(M,r) \;\le\; 2^{17}\ell, \quad \ell = 1 + \ln\frac{M+2}{r}.}$$

The same inequalities hold if the source tables are restricted to the original all-state pair-chain certificate $P^2(u,v) \ge M^{-1}\Pi(v)$, where $P((a,b),(b,c)) = T(b,c)$ and Π is its stationary pair law. The upper covers all tables; stationary lower witnesses already satisfy this restriction.

The certificate is not required for first-order sufficiency. This statement does not remove the certificate from any second-order upper theorem. The lower tests clean and attacked risks of the same map; it is not a logarithmic risk penalty at each individual budget.

Let $R^{\mathcal{T}}_{\star}(M,k;r)$ denote the corresponding minimax risk when k is supplied, using the same observation and source convention.

Proposition 2 (Same-information known-budget scale)**.** *For either source convention and every allowed M, r, k,*

$$\frac{g_M(k)}{96} \le R^{\mathcal{T}}_{\mathrm{stat}}(M,k;r) \le R^{\mathcal{T}}_{\mathrm{all}}(M,k;r) \le 2^{16} g_M(k).$$

The floor holds even with full corrupted history; the upper uses only reconstructed suffix counts and the terminal symbol.

These constants matter. If $D_\star$ is the optimal factor defined as above but normalized at budget k by the *exact* $R^{\mathcal{T}}_{\star}(M,k;r)$, then

$$2^{-16}\Lambda_\star \le D_\star \le 96\Lambda_\star.$$

In particular $2^{-49}\ell \le D_{\mathrm{stat}} \le 96\,2^{17}\ell$ and the same outer bounds hold for D_{all}. No ordering between the two exact-oracle factors is inferred from these separate normalizations.

Corollary 3 (Raw widths and fixed fractions)**.** *For any prescribed integer widths $2 \le r_M \le M+2$, a single budget-blind family has a finite uniform factor over all $M \ge 64$ if and only if*

$$\liminf_{M\to\infty} r_M/M > 0.$$

For $r_M = \max\{2, \lceil \alpha(M+2) \rceil\}$ and fixed $0 < \alpha \le 1$, the optimal uniform factor has order $1 + \ln(1/\alpha)$, with absolute constants. The horizon tends to infinity with α fixed before studying $\alpha \downarrow 0$.

Indeed the finite upper is uniform in M for each fixed fraction, and the finite lower converges to the displayed logarithmic scale. Positive liminf bounds all sufficiently large horizons; the finitely many earlier ones are harmless since loss is at most one and $g_M \ge M^{-1/2}$. If the liminf vanishes, a subsequence makes ℓ diverge. Thus counts plus both endpoint pairs ($r = 2$) cost order $\ln M$. The stronger numerical fixed-fraction lower from the second-order manuscript is *not* transferred to the first-order subclass.

3 The corrected upper interface

Discard X_0. An arbitrary initial pair cannot be treated as an ordinary first-order initial edge. For example, the identity table with initial pair 01 generates $0111\cdots$. The homogeneous suffix $X_1, \ldots, X_{M+1}$ has exactly M actual transitions. On every corrupted word, its ordinary pair counts satisfy

$$N'_{ab} = \sum_{c=0}^{1} H_{abc}(Z) + \mathbf{1}\{(Z_M, Z_{M+1}) = (a,b)\} - \mathbf{1}\{(Z_0, Z_1) = (a,b)\}. \tag{1}$$

Both endpoint pairs are observed because $r \geq 2$. All terms refer to the *corrupted* word. The sum counts edges $0, \ldots, M-1$; adding edge M and subtracting edge 0 leaves edges $1, \ldots, M$. In particular an edit of Z_0 cancels exactly. These nonnegative counts sum to M.

Conditioning is proof-only. For each fixed clean $X_0 = b$ of positive probability, the suffix starts with an arbitrary conditional law of X_1 and thereafter follows T. Define K_b on *every* suffix word by prepending the fixed bit b, applying K, and deleting its first output coordinate. This is a fixed source-independent suffix kernel. It changes at most k suffix symbols and protects the terminal symbol (indeed both final symbols). It is legal in the larger one-protected-symbol attack class of the natural first-order upper. The same predictor is applied for both values of b; uniformity in the initial law and kernel permits averaging the two conditional bounds. Neither b nor any clean symbol is supplied to the predictor.

Suffix-only access is essential. Put $r' = \min(r, M+1)$ and $m = \min\{r'-1, \lfloor M/2 \rfloor\}$. The actual last r' symbols of the suffix are available. Its first r' symbols need not be: they could require the unavailable original symbol Z_r. The accepted upper uses only counts and the last $m+1$ symbols, so this missing prefix is irrelevant. The suffix exposes m ordinary departures, not the pair-state formula $r-2$.

For the selected terminal symbol $u = Z_{M+1}$, let $J'_u = \sum_b N'_{ub}$, $C'_u = N'_{u,1-u}$, and let O'_u be J'_u minus the number of departures from u among the last m suffix edges. Use

$$\widehat{q} = \begin{cases} C'_u/J'_u, & J'_u > 0 \text{ and } 512 O'_u \geq J'_u, \\ 0, & \text{otherwise.} \end{cases}$$

This defines one measurable budget-blind map on the common observation. The natural first-order bound at horizon M gives $2^{17}\ell' g_M$, where $\ell' = 1 + \ln((M+1)/r') \leq \ell$ (including $r = M+2$, when both equal one).

Imported analytic interface. For $a = T(0,1) > 0$ and $t = T(1,0) > 0$, stationary weights $p = (t,a)/(a+t)$ and $h = (B/t, B/a)$, $B = \max(1, a+t)$, satisfy $\sum_u p_u q_u h_u = B \leq 2$. Generic occupation, empirical row-stack, reciprocal-return and old-visit imbalance lemmas then yield the gate bound. Reducible boundaries use direct two-state first-hit control; periodic tables are allowed. The known-budget count estimator is $(C' - 6k)_+/(J' + 6k + 2)$ for $k \geq 1$, and $(C'+1)/(J'+2)$ for $k = 0$, giving $2^{16} g_M$. This paragraph identifies dependencies, not a replacement proof of their universal lemmas.

4 Lower interface and independent oracle floor

The richer-count lower is a separate first-order derivation, not a consequence of the natural pair-count lower. It uses $N = M + 1$ ordinary edges on the same $M + 2 = N + 1$ symbols. With stationary alternatives $T_a(0, 1) = a$, $T_a(1, 0) = t$, set ordinary counts $v = N_{00}$, $u = N_{01}$, and $F_1 = \{X_0 = 1, X_N = 0\}$. The exact skeleton identity is

$$Q_a(F_1, v, u) = \pi_1(a)\binom{v+u}{u}a^u(1-a)^v h_{N-v-u,u}, \quad h_{H,u} = t\binom{H-1}{u}t^u(1-t)^{H-1-u}.$$

Infeasible indices have zero mass. The stationary factor is retained in the equality. For $u \geq 1$ the beta integral is bounded by $1/u$. At fixed v, the remaining sum is $tR_{N-v-1} \leq t$, since $R_0 = 1$, $R_1 = 1 - t$ and $R_n = (1 - t)R_{n-1} + tR_{n-2}$. Thus fixed-$v$ mixture mass with $u \geq u_0$ is at most t/u_0. On the stronger endpoint event $F = \{X_0X_1 = 11, X_{N-1}X_N = 00\}$, triple-departure occupation is $j = J_{00} = v - 1$; it is not v.

The accepted triple-preserving interior relocation freezes both windows, moves only 111 and 000 loops to ordered retained $11, 00$ visits, and restores the interior length. On its good event it preserves all eight counts and both alternative windows, with Hamming distance at most $3J_{\text{cross}} + 4r - 5$ from $1^{N+1-j}0^j$. That null word has $J_{00} = j - 2$, not j; matching is between the *output* and sampled alternative, not the null input. A predeclared mixture proposes once; actual Hamming and final-pair checks accept or return the input unchanged. All other inputs also use identity. This produces one legal strict-budget kernel, with no resampling and no true-source input.

Good-event control includes a separate ordinary-00 count condition, endpoint-conditioned coupling, and the forced-edge discrepancy $1+\sigma$. The risk argument discards nonnegative losses before reducing mixture coefficients, and only then uses the signed triangle bound. Its finite contradiction uses the same map's clean risk and a chosen legal integer attacked budget. It proves, for every richer-observation map, a factor at least $2^{-32}\ell$ relative to g_N, even within the all-state certificate $1/(N - 1)$ class.

Pay the horizon conversion. Pointwise at the same budget,

$$g_N \leq g_M \leq \frac{M+1}{M}g_N, \qquad \frac{R}{g_M} \geq \frac{M}{M+1}\frac{R}{g_N}.$$

Take suprema, then the infimum over maps: the lower is at least $(M/(M + 1))2^{-32}\ell \geq 2^{-33}\ell$. All budgets $0, \ldots, N - 1$ become precisely $0, \ldots, M$. No substitution $N = M$ or attainment is used.

The oracle floor has its own IID proof. Stationary IID Bernoulli sources lie in the first-order subclass and have $P^2(u, \cdot) = \Pi$. Clean two-point testing gives at least $1/(64\sqrt{M})$. For $k \geq 1$, bucket the ones among the first M symbols in multiples of $k+1$ and uniformly retain that many ones, preserving the final pair. This edits at most k bits on every input. Conditional on retained size and the final pair, the arrangement is parameter-independent. There are at most $4(\lfloor M/(k+1)\rfloor+1)$ labels, *including all four protected labels.* The frozen-seed covering argument yields at least $k/(96M)$. Combining the two witnesses gives $g_M/96$ even for full history. This uses the actual IID argument, not a restriction of an arbitrary second-order lower.

5 Exact message access and the second-order extension

Corollary 4 (Two-symbol supplemental alphabet)**.** *In a separate scan-and-message model, the decoder receives* only *the $r = 2$ baseline (eight triple counts, initial pair and final pair) plus one exact message from a fixed finite alphabet. No growing raw windows or other side channels are supplied. A budget-blind encoder may scan the full already corrupted word. The minimum alphabet size for a finite uniform g_M factor is exactly two, for either stationary or arbitrary initial pairs.*

To see sufficiency, discard Z_0 and write $W_i = Z_{i+1}$ for $0 \le i \le M$. For $u = W_M$, set

$$J'_u = \sum_{i=0}^{M-1} \mathbf{1}\{W_i = u\}, \qquad J'_{u,\mathrm{half}} = \sum_{i=0}^{\lfloor M/2 \rfloor - 1} \mathbf{1}\{W_i = u\}, \qquad B = \mathbf{1}\{512 J'_{u,\mathrm{half}} \ge J'_u\}.$$

The decoder reconstructs counts by (1) and uses C'_u/J'_u only when $J'_u > 0$ and $B = 1$, otherwise zero. Its risk is at most

$$\min\{1, 1024 M^{-1/2} + 2^{20} k/M\}.$$

The same conditional-kernel argument applies to this fixed suffix encoder/decoder. A one-symbol alphabet is constant, leaving the $r = 2$ baseline, whose factor diverges by Theorem 1. This proves necessity only in the stated baseline model. Scanning and exact counters are uncharged message-side resources; the bit is fault-free, not a sampled raw coordinate, noisy bit, or bounded-precision implementation.

An assumption-bearing extension, not an order-only phase diagram

Both rows below use the original $M + 2$ word, M triple departures, strict edits protecting the final pair, eight exact triple counts and both raw windows. Factors are relative to g_M; $\ell = 1 + \ln((M+2)/r)$.

Transition class	Assumption for upper	Accepted finite factor range
Binary first-order table	None; arbitrary initial pair	$2^{-33}\ell \le \Lambda_{\mathrm{stat}} \le \Lambda_{\mathrm{all}} \le 2^{17}\ell$
Binary second-order table	All-state $P^2(u,v) \ge \Pi(v)/M$; arbitrary initial pair	$2^{-30}\ell \le \Gamma_{\mathrm{stat}} \le \Gamma_{\mathrm{all}} \le 2^{20}\ell$

The second-order statement is retained from the accepted finite manuscript, including its complete supplement and review closure. It adds source coverage under a supplied certificate; the first-order lower already establishes the qualitative positional obstruction. Its stronger fixed-fraction numerical lower remains within that theorem's own source class. Its separate message theorem has its own access scope, not an inherited removal of the initial-pair input here.

The accepted natural first-order experiment instead has $N + 1$ symbols, N ordinary departures, four pair counts and one protected terminal symbol. Its factor range $2^{-30}[1 + \ln((N+1)/r)]$ to $2^{17}[1 + \ln((N+1)/r)]$ supports the upper machinery but is not itself the richer-count lower. Full-history joint budget/certificate adaptation and unknown-order questions are excluded from this addendum's dependency chain.

6 Proof map, reading limits, and status

This is a consolidation interface, not a standalone complete proof supplement. Reader-facing statements above are bound to exact proof, review and acceptance files by `SOURCE_MAP.json`; historical "candidate" headings do not override later receipts. The required proof closure is:

1. **Generic positive lemmas:** universal occupation, row-stack empirical error, first-hit/return trace, endpoint and occupation tails, weighted half-imbalance, old-prefix reciprocal, weighted minimum, and deterministic gate perturbation. Their full proofs are in the accepted positive supplement. The natural first-order upper specializes them to two states, treats reducible boundaries separately, and proves the exact half-visit-bit bound.
2. **Richer-observation lower:** the independent first-order fixed-v skeleton and renewal mixture, conditional endpoint good event with separate 00 control, all-row certificate check, legal single-proposal kernel, and finite clean-versus-attacked contradiction. The deterministic triple/window relocation is an accepted separate lemma; the natural pair-count lower is contextual only, not a replacement for this derivation.
3. **Common-model interface:** corrupted-count reconstruction, discarded-first-edge correction, induced fixed-b kernel, suffix-only measurability, paid $N = M + 1$ lower conversion, and the independent protected-two-symbol IID oracle floor. These are the matching steps detailed on pages 3–4.
4. **Extension closure:** the accepted finite second-order manuscript's positive, negative and finite-horizon supplements remain required for its table row. Their source-specific four-state arguments are not imported into the no-certificate first-order upper.

A future standalone supplement would have to reproduce the full proofs in items 1–3 (and item 4 if retaining the extension), not merely cite this map. No new theorem branch, neural realization, full-history impossibility, or joint order-adaptation claim is added.

Closest-work comparison, inherited bounded read scope

The following comparisons inherit the local primary-source audits and their version/read-scope ledgers. This addendum did not reread every external paper. No inspected subsumption was established by those audits; that is not an exhaustive novelty or priority conclusion.

- **Han–Jana–Wu (2023):** actual-terminal prediction and suffix methods are direct antecedents. The journal's squared-TV discussion is not an unsquared-TV theorem for this edited, restricted-observation experiment; the binary squared-loss open question is not resolved here. See journal DOI, model (1) and section VI.A as bounded in the audit.
- **Jain–Orlitsky–Ravindrakumar and Bhatt et al.:** unknown-contamination adaptation is established prior work. The former's geometric selection needs simultaneous validity of all appropriate balls; the latter supplies confidence intersection in an IID mean model. Separate expected risks here do not supply that premise. A random terminal target alone is *not* a barrier to their pointwise logic. Audited scopes: 2022 note, sections 2–3; ICML 2022 paper, section 6/Appendix A.7.
- **Robust counts and Markov types:** Ghosh's transition-mixture parametric inference and Leskelä–Dreveton's parallel-trajectory robust count estimator differ in attack, target and guarantee. They preclude broad anti-count claims. Markov exchangeability, finite stacks and constrained-risk arguments are antecedents; bounded-Hamming matching, protected boundaries and mixture control still require the proofs listed above. Exact versions and reading limits are in the inherited audit ledgers.

Status boundary. Underlying units have local root acceptance after independent LLM review; the current integration status of this assembled addendum is recorded separately in `STATUS.md` and its receipt. Build/render checks are separate evidence, not scientific acceptance, formal verification, external peer review or publication permission. Accepted source files remain unchanged. No authorship, affiliation, license, novelty, learning, or Transformer-performance claim is introduced.